

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

CONTROL SYSTEMS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Assume suitable data and illustrate answers with sketches wherever necessary.
- (4) Use graph paper and semilog paper wherever necessary.

1. (a) Using block diagram reduction technique for the system shown in Fig. 1 below, determine the overall transfer function $C(s)/R(s)$.

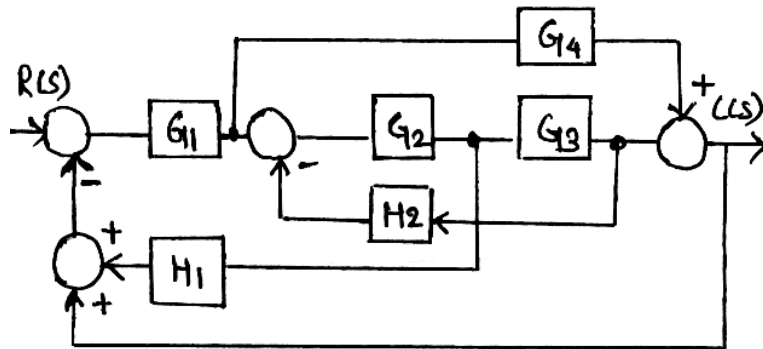


Fig. 1

6(CO2)

- (b) Determine the overall transfer function using Mason's Gain formula for Fig. 2 given below.

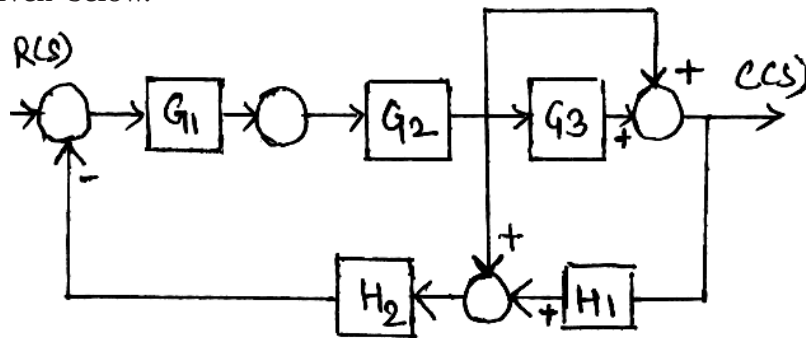


Fig. 2

6(CO2)

2. (a) The forward path gain of a unity feedback control system is

$$G(S) = \frac{144}{S(S + 12)}$$

Determine the natural frequency, frequency of damped oscillations, maximum overshoot, peak time, rise time, settling time and the expression for the unit step response. 6(CO2,4)

- (b) A unity feedback system has an open loop gain

$$G(S) = \frac{K}{S(S^2 + 4S + 20)}$$

Specify the type of the system. Evaluate the static error constants and the corresponding steady state error. Assume that the system is stable. 6(CO4)

3. (a) A unity feedback control system has an open loop transfer function

$$G(S) = \frac{K(S + 13)}{S(S + 3)(S + 7)}$$

Using the Routh-Hurwitz stability criterion, find the range of K for which the system to be stable. Check all poles of the closed loop transfer function having damping factor greater than 0.5. Assume unity feedback. 6(CO3)

- (b) Sketch the root locus for a system with open loop transfer function as,

$$G(S)H(S) = \frac{K(S + 5)}{S(S^2 + 4S + 5)} \quad 6(CO3)$$

4. (a) Sketch the polar plot for

$$G(S) = \frac{32}{(S + 4)(S^2 + 4S + 8)}$$

Find its points of intersection with the real and imaginary axes. 6(CO3,4)

- (b) Sketch Bode plot for a unity negative feedback system given by :

$$G(S) = \frac{12650}{(s + 10)(S + 20)^2} \quad 6(CO4)$$

5. (a) For the electrical network shown in Fig. 5 (a), select a set of proper state variables and write down a state equation in physical-variable form, to represent the system.

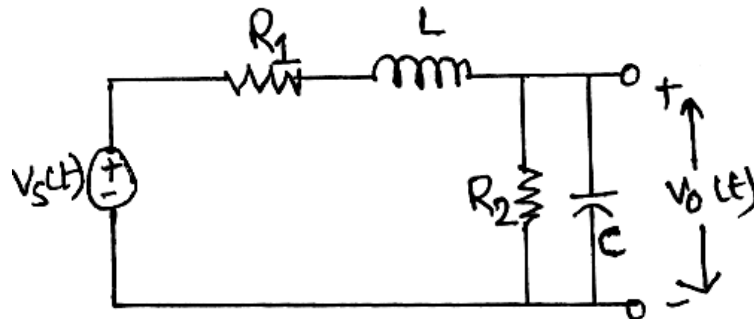


Fig. 5 (a)

6(CO5)

- (b) What is saturation non linearity ? 3(CO1)

- (c) State the advantages and disadvantages of Optimal control system.

3(CO1)

